Combinatorial problems with analytic flavor

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Certainly the best times were when I was alone with mathematics, free of ambition and pretense, and indifferent to the world [1]

1 Crash course in Analysis

Let us review a few useful definition and theorems from real analysis.

Definition 1.1 (open and closed subset). Let $X \subset \mathbb{R}^n$, we Say that X is closed of the limit of every convergent sequence $(x_n), x_i \in X$ lies also in X. We say that X is open if the complement of X is closed. In other words, X is open if for every $x \in X$, there is $\epsilon \in X$ such that

$$\{y \in \mathbb{R}^n | ||y - x|| < \epsilon\}$$

Proposition 1.2
A function $f : \mathbb{R}^n \to \mathbb{R}^m$ is continuous iff for every closed(or open) subset $X \subset \mathbb{R}^m$ we have $f^{-1}(X) := \{y \in \mathbb{R}^n | f(y) \in X\}$ is closed(or open).

Definition 1.3. Let $X \subset \mathbb{R}^n$, we say that X is compact if for every sequence $(x_n) \in X$, there exist a subsequence x_{n_i} such that x_{n_i} is also convergent with the limit in X.

Proposition 1.4
X is compact iff it is closed and bounded.

The idea of the proof: reduce to the case of $X \subset \mathbb{R}$ and note that every sequence of real numbers has either an increasing subsequence or a decreasing one.

Proposition 1.5
If $f : X \to \mathbb{R}^n$ is a continuous map and X is compact then the image of f is also compact.

The idea of the proof: let $(y_n) \in f(X)$ take $(x_n) \in X$ such that $f(x_n) = y_n$ and use that X is compact.

[1] Langlands

Corollary 1.6
If $f : X \to \mathbb{R}$ is continuous and X is compact, then f have a maximum and a minimum.

As an exercise, try to prove this fact:

Corollary 1.7
If $X, Y \subset \mathbb{R}^n$ are such that X is compact and Y is closed , and $X \cap Y = \emptyset$ then $d(X,Y) := \{\inf_{x\in X, y\in Y} ||x - y||\}$ is not zero.

> **Remark 1.8.** Must of the this section can be generalized to any space that we have a notion of distance between points, these spaces are called metric spaces.

2 Compact problems

Example 2.1 (Iran training 2025)
There are finitely many segments in the plane such that no lattice point lie on these segments. Prove that there exists a polynomial with two variables that is positive on each segment and negative on each lattice point.

hint: Think about degree 2 polynomials and use 1.7

Example 2.2 (Iran training 2025)
Let $X \subset \mathbb{R}^{\times}$ be a closed convex subset. A corner point of X is a point $x \in X$ such that if

$$x = \sum w_i y_i, y_i \in X, w_i > 0, \sum w_i = 1$$

then $\forall i, y_i = x$. prove that every point in X is of the form $\sum w_i y_i$ where y_i are corner points of X and $\sum w_i = 1, w_i > 0$.

hint let $Y \subset X$ be the convex combinations of corner points, consider the set $Z \subset X$ of the points with maximum distance to Y, fix $z \in Z$ and consider $W \subset Z$ of the points with the maximum distance to z.

> **Remark 2.3.** This is true in every normed vector space, if you know what this means, try to use it to prove that every function $f : \mathbb{Z}^n \to [0, 1]$ such that
>
> $$f(x) = \frac{\sum_{\textbf{y is a neighbor of x}} f(y)}{2^{n+1}}$$
>
> is constant.

Problem 2.4 (Iran 2004). An arbitrary n-gon is places on the plane. its sides are numbered from $1, \ldots, n$. If $S = s_1, s_2, \ldots$ is a sequence(that can be finite or infinite) where $s_i \in \{1, 2 \ldots, n\}$ we move P according to S by first rolling it by side s_1 then rolling it by side s_2 and so on.

- Show that there is a sequence S so that when we roll P according to that sequence evey point of the plane will be covered eventually.
- Show that this sequence can not be periodic.
- Let P be a pentagon inscribed in the unit circle, and D be a circle with the radius 1.0001 on the plane. Is there a finite sequence such that after rolling P according to it, the pentagon be contained entirely in D.

Problem 2.5. (i) Let X and Y be subsets of the vertex set of a regular n-gon. Show that there is a rotation ρ of this n-gon such that $|X \cap \rho(Y)| \geq |X||Y|/n$.

(ii) Let $S = F_1 \cup F_2$, where F_1 and F_2 are closed subsets of S, the unit circle in $\mathbb{R}^2$. Then for some j, $1 \leq j \leq 2$, and every angle $0 \leq \alpha \leq \pi$, the set F_j contains two points x and y, with y obtained from x by rotating it through α.

3 exercises

Problem 3.1 (Iran 2000). Is there any function $f : S^2 \to S^1$ such that for any $x, y \in S^2$ we have $||f(x) - f(y)|| \geq ||x - y||$

Problem 3.2 (Iran 2001). Let $n = 2^m + 1$ and $f_1, f_2, \ldots, f_n : [0,1] \to [0,1]$ be increasing functions which satisfy $|f(x) - f(y)| \leq |x - y| \, \forall x, y \in [0,1] \; 1 \leq x, y \leq n$. Prove that there exists $i \neq j$ such that $|f_i(x) - f_j(x)| \leq \frac{1}{m}$.

Problem 3.3 (Iran 2004). A luminous point is in the space. Is it possible to prevent its luminosity with a finite number of spheres of the same size?

Problem 3.4 (Iran 2005). Suppose that there are 18 lighthouses in the sea, each lighthouses lighten an angle of degree 20. prove that we can choose the directions of the lighthouses such that the whole see lighten, regardless of the place of the lighthouses.

Problem 3.5 (Iran 2006). Suppose that We have a simple polygon(not necessarily convex). show that this polygon has a diagonal which lies entirely inside the polygon and partition the polygon into two sides, each has at least one third of the vertices of the polygon.

Problem 3.6 (Iran 2009). Five research stations are operating on Mars. Is it always possible to divide the surface of Mars into five connected areas, all congruent to each other, such that each area contains exactly one station? Note that stations are points and Mars is assumed to be a complete sphere.

Problem 3.7 (Iran 2011). A simple graph is called self-intersecting iff it can be drawn in the 2-dimensional plane, drawing the edges as line segments and vertices as points, in such a way that each two edges intersect. Note that no edge can pass through a vertex except its endpoints. Find all self intersecting graphs.

Problem 3.8 (Iran 2011). (a) Let $RBR'B'$ be a convex quadrilateral in the 2-dimensional plane. We have colored the vertices R, R' red, and B, B' blue. There are some points strictly inside the quadrilateral. Among all of the points (including the quadrilateral vertices), no four are on a circle, except for possibly the four original quadrilateral vertices. The inner points are arbitrarily colored red or blue. Prove that exactly one of the following situations happens:

- There is a path from R to R', such that for every point on it, the nearest red point is strictly closer than the nearest blue point.
- There is a path from B to B', such that for every point on it, the nearest blue point is strictly closer than the nearest red point.

A path of the first kind is called a red path, and path of the second kind is called a blue path.

Let n be natural number. Two persons are playing a game of $2n$ turns. They alternate turns, putting red and blue points inside the quadrilateral, such that the mentioned constraints (no four points being on a circle) still hold. The first person puts only red points and tries to make a red path appear at the end. The other person puts only blue points and tries to make a blue path appear at the end.

(b) Prove that if $RBR'B'$ is a rectangle, no matter what n is, the second player can always win.

(c) Try to find out which player can win, for other types of quadrilaterals.

Problem 3.9 (Iran 2011). Suppose S be a figure in the plane that its border does not contain any lattice point and let $x, y \in S$ be two lattice points with unit distance from one another. A copy of S is a combination of rotations and reflections of S. Assume that there exists a tiling of the plane with copies of S such that x and y lie on lattice points for every copy of S in this tiling. Prove that the area of S is equal to the number of lattice points in it.

Problem 3.10 (239 2010). You are given a convex polygon with perimeter $24\sqrt{3} + 4\pi$. If there exists a pair of points dividing the perimeter in half such that the distance between them is equal to 24, Prove that there exists a pair of points dividing the perimeter in half such that the distance between them does not exceed 12

Problem 3.11 (Miklos Schweitzer 1982). Show that for any natural number n and any real number $d > 3^n/(3^n - 1)$, one can find a covering of the unit square with n homothetic triangles with area of the union less than d.

Problem 3.12 (ELMOSL 2018). A windmill is a closed line segment of unit length with a distinguished endpoint, the pivot. Let S be a finite set of n points such that the distance between any two points of S is greater than c. A configuration of n windmills is admissible if no two windmills intersect and each point of S is used exactly once as a pivot.

An admissible configuration of windmills is initially given to Geoff in the plane. In one operation Geoff can rotate any windmill around its pivot, either clockwise or counterclockwise and by any amount, as long as no two windmills intersect during the process. Show that Geoff can reach any other admissible configuration in finitely many operations, where

(i) $c = \sqrt{3}$,

(ii) $c = \sqrt{2}$

Problem 3.13 (IMOSL 2012). There are given 2^{500} points on a circle labeled $1, 2, \ldots, 2^{500}$ in some order. Prove that one can choose 100 pairwise disjoint chords joining some of theses points so that the 100 sums of the pairs of numbers at the endpoints of the chosen chord are equal.

Problem 3.14 (Saint Petersburg 2000). In the plane is given a convex n-side polygon $\mathcal{P}$ with area less than 1. For each point X in the plane, let $F(X)$ denote the area of the union of all segments joining X to points of $\mathcal{P}$. Show that the set of points X such that $F(X) = 1$ is a convex polygon with at most $2n$ sides.

Problem 3.15 (Czech and Slovak 2000)**.** In the plane are given 2000 congruent triangles of area 1, which are images of a single triangle under different translations. Each of these triangles contains the centroids of all the others. Show that the area of the union of these triangles is less than $\frac{22}{9}$.

Problem 3.16 (Romania 2000)**.** Let $P_1P_2\cdots P_n$ be a convex polygon in the plane. Assume that for any pair of vertices P_i, P_j, there exists a vertex V of the polygon such that $\angle P_iVP_j = \pi/3$. Show that $n = 3$.

Problem 3.17. A lion and a Christian in a closed circular Roman arena have equal maximum speeds. Can the lion catch the Christian in finite time?

Problem 3.18. Let $\ell(B) = a + b + c$ be the linear measure of an $a \times b \times c$ box (rectangular parallelepiped). This linear measure is monotone in the sense that if a box B is contained in a box C then $\ell(B) \leq \ell(C)$.